Georgia Institute of Technology

Georgia Institute of Technology
College of Computing
School of Computing Instruction
Course: CS 2340 – Objects and Design
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Head TA: Mr. Elias Lind

Team-Based* (same team as the second project) *and Take-Home
[Each part will pose a different AI allowance - **Read our syllabus]**

Midterm Exam – Object-Oriented *Analysis* and *Design + Web Development*

Description

This take-home midterm is designed to assess your understanding of **software engineering principles** covered in CS 2340, as well as your ability to **apply those principles in realistic, professional scenarios**.

Rather than focusing on memorization, this exam emphasizes:

- Conceptual understanding
- Clear technical communication
- Thoughtful application of software engineering practices
- Collaboration and design reasoning

Unless otherwise stated:

- You may reference course materials and your own project work.
- You should answer questions **in your own words**.
- Clear, well-structured explanations are valued more than length.

The goal of this midterm exam is not just to evaluate what you know today, but how well you can **reason, communicate, and think like a software engineer**.

Submission Deadline: Sunday, March 8th, 11:59 PM. Late policy applies according to our syllabus.

Grading

The exam weights will be distributed as follows:

- First part – 20%
- Second part – 35%

- Third part – 30%
  - ***Partial functionality will not be accepted***--*all user stories must be fully functional and respond correctly to their completion steps.*
    Disclaimer: *For full functionality, the user story must work on PythonAnywhere.*
- *Fourth part - 15 %*

Disclaimer No. 1: Your grade will be based on the correctness and clarity of your submission.

Disclaimer No 2: All team members must substantially contribute to the exam and its deliverables. Subpar contributions will be penalized with a grade of zero (0).

Submission

On Canvas, you will have to submit your work as a team (one submission per group). There are ***12 deliverables total***, split into four parts.

Part one, part two, part three, part four

1. **Software Engineering Foundations & Professional Reasoning Video** – .txt file with a sharable link (e.g. Teams, OneDrive, Google Drive, and YouTube)
   a. File must be named "<Your group ID>_foundations.txt"
2. **Domain Model Diagram** for question 4 – Image (PNG or JPEG)
   a. File must be named "<Your group ID>_question_4_DMD" with the appropriate file extension
3. **Use Case Diagram** for question 5 – Image (PNG or JPEG)
   a. File must be named "<Your group ID>_question_5_UCD" with the appropriate file extension
4. **System Sequence Diagram** for question 6 – Image (PNG or JPEG)
   a. File must be named "<Your group ID>_question_6_SSD" with the appropriate file extension
5. **System Sequence Diagram** for question 7 – Image (PNG or JPEG)
   a. File must be named "<Your group ID>_question_7_SSD" with the appropriate file extension
6. **Design Class Diagram** for question 8 (part a) – Image (PNG or JPEG)
   a. File must be named "<Your group ID>_question_8_DCD" with the appropriate file extension
7. **Rationale** for question 8 (part b) – PDF
   a. File must be named "<Your group ID>_question_8_rationale.pdf"
8. **Sequence Diagram** for question 9 – Image (PNG or JPEG)
   a. File must be named "<Your group ID>_question_9_SD" with the appropriate file extension
9. **Sequence Diagram** for question 10 – Image (PNG or JPEG)
   a. File must be named "<Your group ID>_question_10_SD" with the appropriate file extension

10. **Analysis and Design of an Object-Oriented System Video** – .txt file with a sharable link (resources such as Teams, OneDrive, Google Drive, and YouTube are available)
    a. File must be named "<Your group ID>_analysis_and_design.txt"

11. **Implementation, Deployment, and Version Control** – .txt file including:
    a. Process video
    b. GitHub repository
    c. PythonAnywhere link to deployed project
    d. ***Note***: the file must be named "<Your group ID>_implementation_and_deployment.txt"

12. **Project Three Ideation and Early Design**– .pptx file (slides only) including:
    a. End User Persona description.
    b. Custom user stories (two).
    c. Wireflows (two–one per custom user story).

Details

This section will go into the specifics of what is expected from each section; please note that the scenarios and question specifics can be found in the next section.

1. **Part one**: one deliverable
    a. **Video podcast:** With all members present, explain your responses in dialogue-based, explanatory, and interactive way.
        i. Your team will meet in person or remote, as if you all were hosts of a podcast (communication skills are essential when applying to jobs; use this task to help your skills grow).
        ii. All of your faces must be present in the video (using video recording, not just a picture).
        iii. Your video must be **at most 21 minutes** (It can be shorter, but no longer). We suggest ~7 minutes per question.
        iv. Refer to the "Appendix" section to further understand what is expected from the podcast.

2. **Part two**: **nine** deliverables
    a. All diagrams must be created digitally; you **must** use Draw.io
    a. Please see the questions section for more specifics on the diagrams.
    b. **Video Presentation:** You will submit an "***interview***" video, as a team, to present and explain all diagrams you created for this part.
        i. Your video must be **at most 24 minutes** (It can be shorter, but no longer). We suggest ~3 minutes per diagram.
        ii. All of your faces must be present in the video (using video recording, not just a picture).
        iii. Refer to the "Appendix" section to further understand what is expected from the "interview" video.

3. **Part three:** one deliverable
    a. **Partial functionality will not be accepted** – all user stories must be fully functional and respond correctly to their completion steps. Your team must submit **one** deliverable with three items.
    b. **.txt file** titled "<Your group ID>_implementation_and_deployment.txt" including links to:
        - **Process video:** The video recording must showcase your **full development and deployment process from start to finish**. That includes all interactions and contributions to the team's repository. **Each team member** is responsible for recording their own work on the user stories (one per team member) they are assigned. These individual recordings should then be assembled into a **single video** deliverable (PowerPoint is a useful tool for editing and assembling the final video).
        - **GitHub Repository:** The team must grant access to their client TA.
        - **PythonAnywhere** link to the deployed, fully functional project.

4. **Part four:** one deliverable– a .pptx file (slides only) including:
    a. End User Persona description.
    b. Custom user stories (two).
    c. Wireflows (two–one per custom user story).

Questions

Part One: Software Engineering Foundations & Professional Reasoning
****AI limited****

Question 1: Describe the Model-View-Controller (MVC) architectural pattern and explain how (and why) Django relates or could be based around the Model-View-Controller architecture. You may use your project's code as an example to elaborate on your explanation.

Question 2: After graduating from Georgia Tech, you are hired as a project manager for a software company that builds and maintains Django web applications for clients. The company has been around since the 1990s and is still using Waterfall for project management. While talking to clients, you have noticed that they consistently express frustration with the fact they are unable to request changes during the development process without causing significant delays. After recalling the lessons you learned in CS 2340, you realize Agile would be a better fit for your company.

Imagine you are meeting with your supervisory team. Present and explain the benefits of swapping over to an Agile methodology for project management (Hint: you will need to be convincing).

Question 3: Explain the role of collaboration in a software engineering process and elaborate on why using Git/GitHub could be helpful (Keep in mind that you will need to be convincing!).

Part Two: Analysis and Design of an Object-Oriented System
****AI limited****

Context for Analysis and Design

Georgia Tech Yellow Cross has hired you to help design a nationwide system to manage its hospitals, patients, and physicians.

The company (which has a name) has multiple hospitals, each with an address, an ID, and a name, and each with multiple physicians (zero or many). Physicians can only work at one hospital at a time. Among physicians, hospitals can have attendings, residents, and interns. All physicians have a unique ID. An attending generally has a name, a surname, a specialty, a salary, a date of birth, a certification date (when they became an attending), and certified experience (measured in years). A resident has a name, a surname, a specialty, a salary, a date of birth, and a residence start date. An intern has a name, a surname, a date of birth, and a salary.

Attendings generally have one to three residents under their charge. However, on a given day, they work with only one of their cases (i.e., current residents). On the other hand, a resident has up to five interns responding to them. However, when a resident is sanctioned, all their interns are assigned to someone else.

Physicians in a hospital belong to departments (e.g., Neurosurgery)--each with a name. A physician can only belong to one department at a time. Each department has a department chief, and each hospital has a doctor serving as their general chief.

The company supports many patients nationwide. A patient has a name, a surname, a date of birth, a unique ID, and an affiliation type. Patients can have one to three preferred hospitals. Also, the company generally manages patients and their clinical cases. A case has a title, a description, a start date, an end date (that could be undefined), and an ID; it generally indicates the patient, all hospitals involved, and all attendings who participated.

Scenarios

Scenario No 1: *Meredith, a managing member of GT Yellow Cross, wants to generate a historical report of all existing clinical cases in the company for her upcoming meeting. She logs into the system, which first verifies her credentials through an external authentication system (i.e., LDAP). Once logged in, she requests the system to list all clinical cases and sort them based on their start date. The system then accesses an external database system that outputs all data needed for the report. Shen then proceeded to ask the system to generate a*

PDF, which she later sent to a company's printer to get it physically. Once done, Meredith logs off and heads to her meeting.

Scenario No 2: *Derek, a neurosurgery attending, is terrible with names and a shy human being. Before he goes through his rounds, he needs to figure out the names of the interns that will be in his cases during the day. He first logs into the system, inputting his credentials, and looks for his name. Once he finds himself, he asks the system to indicate the resident working for him during the day. He then asks the system to list the names of all interns under his current resident's charge. Once done, Derek logs off and heads to his rounds.*

Question 4: Considering the previous context, create a domain model diagram that represents the context, its actors, relationships, and attributes.

Questions 5: Considering the previous context and its two scenarios, draw a use case diagram that includes:

- Two or more actors.
- At least five use cases.
- At least one <<extends>> relationship.
- At least one <<include>> relationship.

Question 6: Draw a system sequence diagram to illustrate the first scenario.

Question 7: Draw a system sequence diagram to illustrate the second scenario.

Question 8:

- Part A: Considering the previous context and its two scenarios, draw a design class diagram.
- Part B: Describe your rationale based on Design Principles (SOLID/GRASP) and good practices of software design.

Question 9: Draw a sequence diagram to illustrate the first scenario, following the structure and design of your DCD.

Question 10: Draw a sequence diagram to illustrate the second scenario, following the structure and design of your DCD.

Part Three: Implementation, Deployment, and Version Control
****AI permitted****

For this third part, your team must implement five new user stories for the GT Movies Store project and submit a txt file titled "<Your group ID>_part3.txt." (See submission details in this document). One of the three items to submit is a link to a single video recording that showcases your development and deployment process from start to finish: that is, you record during the process, not afterward!

Each team member is responsible for recording their own work (one user story per member), including both screen and audio, while describing and reflecting on their live process. That includes all interactions and contributions to the team's repository. These individual recordings must then be combined into one cohesive team video; PowerPoint is a useful tool for editing and assembling the final deliverable. All team members are required to contribute, and the use of open notes is permitted.

Your implementation and deployment process must be fully documented (i.e., recorded live) in the recording. In addition, all user stories must be successfully deployed on PythonAnywhere in order to count toward your grade. **Partial functionality will not be accepted**--all user stories must be fully functional and respond correctly to their completion steps.

Please note: As a team, choose one Project 1 codebase (from any team member) and use it as your foundation. You will collaboratively implement all required extensions on that single codebase on a shared repository. All development activities (including your team's work, progress updates, and extensions) must be documented and tracked in GitHub.

User Stories:

1. **As a User**, I want to see a map showing which movies of the GT Movie Store are trending (or most purchased) in different geographic areas, so I can discover what's popular around me or in other regions.

   Completion steps:
   a) Log in or register an account.
   b) Navigate to the "Local Popularity Map" page.
   c) Verify that the map loads correctly with regional boundaries or markers.
   d) Confirm that movies with high purchase/view counts are displayed as trending in at least one region.
   e) Select a specific region on the map.
   f) Verify that the region's top trending movies are listed (titles, counts, etc.).
   g) Compare this list with your own recent purchases or another user's purchases to ensure accuracy.

2. **As a User**, I want to rate movies (e.g., 1–5 stars) so I can quickly share feedback without writing a full review.

Completion steps:

a) Log in or register an account.
b) Navigate to a movie's detail page.
c) Select a rating option (e.g., 1–5 stars or thumbs up/down).
d) Submit the rating.
e) Refresh or revisit the movie's detail page.
f) Verify that:
    i. Your rating is saved and visible.
    ii. The movie's average rating updates accordingly.
g) Log out, then log back in to confirm that your rating persists under your account.

3. **As an admin**, I want to see (on a separate tab/screen) which user on my movie store has purchased the most movies and what that number is (movies purchased).

   Completion steps:

   a) Log In to the GT Movie Store
   b) Make a purchase from a regular User account
   c) Navigate to the Admin Dashboard to display the separate tab/screen
       i. This should show that User and their purchases
   d) Switch to another regular User account and make more purchases compared to the first User
   e) Navigate to the Admin Dashboard to display the separate tab/screen
       i. This should show that other User and the number of movies purchased

4. **As an admin,** I want to see (on a separate tab/screen) which user on my movie store has commented the most and what that number is (comments posted).

   Completion steps:

   a) Log in to the movie store as a user.
   b) Make a comment on a movie so that this user has the most comments made.
   c) Navigate and log in as an admin.
   d) Show the separate tab/screen that indicates the user who has commented
   e) the most and then the correct number of comments made.
   f) Navigate back to the movie store as a ***different*** user and comment more times
   g) than the previous user.
   h) Navigate back to the admin page.
   i) Show the separate tab/screen and that it indicates that the second user has
   j) commented more times and the correct number of comments made.

5. **As an admin**, I want to see (on a separate tab/screen) which movie on my movie store has been purchased the most and the movie that is reviewed the most. Your separate tab/screen should display the movie names and the times they've been bought/reviewed, respectively. If one movie fits both criteria, show it twice for the two measurables.

   Completion steps:

   a) Log in to the movie store as a user.
   a) Comment on 1 movie
   b) Purchase the ***same*** movie
   c) Navigate to the tab/screen in admin you created for this User Story
   d) Show that the movie shows up correctly for most reviewed and
   e) most bought
   f) Purchase a different movie twice
   g) Show that the most bought and most reviewed movies are not the
   h) same in your tab/screen in admin

Part Four:
****AI limited****

This fourth part is designed to help you practice and reinforce your understanding and skills in software analysis, design, and implementation. You will help tailor your third project's deliverables. The outcome from this part will determine what your project will be!

Your task for this part:

1. You will read the [third project's context](). Please note that the first sprint for the third project will begin in week 10: you don't have to develop it yet!

2. Based on the proposed backlog, you are required to:

   a. Identify who your End User persona is (who the chatbot is for and what problem it solves). The artifact from this step will be a Persona description as suggested by the NN/Group: [Personas Make Users Memorable - NN/G]()

b. Write down at least **two custom user stories** specific to your chosen domain (a domain could be education, health, sports, etc.).

c. You will follow a user-centered, top-down design approach. Using this approach, you will design the User Interface (UI) and visually illustrate **two navigation schemas** through the UI. Each navigation schema should correspond to one custom user story that reflects the overarching visual design of your end product: a web application hosting your RAG-based chatbot.

You must develop two low-fidelity wireframe-flow diagrams (i.e., wireflows), one per custom user story. Each diagram should clearly show how users navigate through the interface to accomplish their goals, emphasizing the structure, flow, and key interaction points of your design. Useful tools to create wireframes are: Figma, Balsamiq, Draw.io, Free Online Miro, and Lucidchart.

Appendix

How are we defining what an interview is?

An **interview** is structured around **eliciting insights from one or more individuals** with relevant experience or perspective.

The interview format is intended to ensure that your team can **clearly explain and justify your diagrams and design decisions** while visually presenting them on screen.

You should **show your diagrams** and discuss them in an interview-style manner. This does **not** need to follow a rigid or formal structure. Acceptable formats include (but are not limited to):

- One team member asking explicit, interview-style questions about the diagrams (e.g. "Why did you model this association as one-to-many?") while another responds
- Team members prompting each other with prepared questions
- A structured walkthrough where questions guide the explanation

The specific format is flexible. What matters is that diagrams are clearly shown, design decisions are explained aloud, and **all team members actively participate and demonstrate understanding**.

Key characteristics:

- Clear interviewer(s) and interviewee(s)
- Prepared, intentional questions

- Focus on *extracting knowledge*, not debating opinions
- Interviewer(s) guide the conversation

Good interview questions often explore:

- How certain tools, processes, or methodologies are actually used
- Tradeoffs and decision-making in real projects
    - Practical use-cases
    - Comparisons between different diagrams
    - When and *why* certain approaches work or fail

How are we defining what a podcast is?

A **podcast-style discussion** is more conversational and idea-driven. Rather than extracting information from a single expert, the group engages in **open discussion, debate, and reflection**.

Key characteristics:

- More balanced speaking time among group members
- Fluid talking structure (little rigidity)
- Emphasis on discussion, disagreement, and synthesis
    - Present thought-provoking viewpoints or "hot takes"
- Participants build on each other's ideas

Good podcast topics often involve:

- Tradeoffs (e.g., speed vs. learning, automation vs. understanding)
- Evolving industry practices
- Personal experiences and opinions, grounded in course concepts